Evaluating Demographic Differences in Machine Learning–Based Hospital Readmission Prediction

Abstract

This project evaluated whether machine-learning models for 30-day hospital readmission perform consistently across demographic groups using the UCI Diabetes 130-US Hospitals dataset. Logistic Regression and XGBoost models were developed using demographic, clinical, hospitalization, medication, and prior healthcare-utilization variables. To prevent information leakage, the data were split at the patient level so that no patient appeared in both the training and test sets.

XGBoost achieved slightly better overall performance, with an AUROC of 0.637 and recall of 0.525. Subgroup analysis showed relatively small differences by gender, while greater variation was observed across race and age groups. The Asian subgroup had the lowest race-specific recall and was also the least represented racial group. However, a race-based weighting experiment did not consistently improve subgroup performance and slightly reduced overall model performance. These findings suggest that demographic representation may be related to subgroup performance, but representation alone does not fully explain the observed differences.

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1. Introduction

Hospital readmission is an important healthcare outcome because patients who return to the hospital shortly after discharge may require additional clinical attention, follow-up care, or better care coordination. Machine-learning models can support readmission-risk prediction by identifying patterns in patient demographics, hospitalization characteristics, diagnoses, medications, and prior healthcare utilization. However, strong overall model performance does not necessarily mean that a model performs equally well for all patient groups.

This project evaluates demographic differences in machine-learning–based prediction of 30-day hospital readmission using the UCI Diabetes 130-US Hospitals dataset. The dataset contains 101,766 hospital encounters involving 71,518 unique patients and includes information on race, gender, age, diagnoses, medication use, hospitalization characteristics, and previous inpatient, outpatient, and emergency visits. The prediction target was defined as readmission within 30 days, with <30 coded as 1 and >30 or NO coded as 0.

Because some patients appeared in the dataset more than once, the analysis used a patient-level train/test split rather than randomly separating individual encounters. This ensured that no patient appeared in both the training and test datasets and reduced the risk of information leakage. Logistic Regression was used as a baseline model, while XGBoost was used as the nonlinear machine-learning model. Performance was evaluated using AUROC, AUPRC, precision, recall, F1-score, and confusion matrices.

The main focus of the project was to determine whether model performance differed across race, gender, and age groups and whether these differences were related to group representation. Subgroup performance was therefore examined alongside the number of patients and positive readmission cases within each group. A race-based sample-weighting experiment was also performed to test whether giving less-represented racial groups greater influence during training changed subgroup performance. The goal was to identify performance differences without assuming that those differences necessarily represented bias or were caused by representation alone.

1.1 Project Objective

The objective of this project is to evaluate whether a machine-learning model for 30-day hospital readmission performs consistently across demographic groups and to examine how differences in group representation may relate to subgroup model performance.

1.2 Research Questions

1. Can machine-learning models predict 30-day hospital readmission?
2. Does model performance differ across race, gender, and age groups?
3. Are subgroup performance differences associated with differences in demographic representation?
4. Does race-based sample weighting reduce observed subgroup performance differences?

2. Dataset and Data Structure

The dataset used in this project was the Diabetes 130-US Hospitals for Years 1999–2008 dataset from the UCI Machine Learning Repository. It contained 101,766 hospital encounters and 50 variables involving 71,518 unique patients. The data included demographic characteristics such as race, gender, and age; hospitalization information including admission type, time in hospital, laboratory procedures, medications, and number of diagnoses; prior healthcare utilization through inpatient, outpatient, and emergency visits; diagnosis codes; medication-related variables such as insulin use, diabetes medication status, and medication changes; and the hospital readmission outcome.

Two identifiers were especially important: encounter_id, which uniquely identified each hospital encounter, and patient_nbr, which identified individual patients. No duplicate encounter IDs or fully duplicated rows were found, but 16,773 patients had repeated encounters, so patient_nbr was used to create a patient-level train/test split and was excluded from the model predictors. The original readmitted variable contained <30, >30, and NO; for this project, it was converted into a binary target called readmit_30, where <30 was coded as 1 and both >30 and NO were coded as 0. This produced 11,357 positive 30-day readmissions and 90,409 non-positive encounters, making readmission a relatively uncommon outcome in the dataset.

2.2 Data Relationship

The main relationship in the dataset can be summarized as:

Patient → Hospital Encounter → Readmission Outcome

A patient may have one or multiple encounters, while each encounter contains its own hospitalization, diagnosis, medication, and readmission information. This structure was especially important when creating the training and test datasets because encounters from the same patient needed to remain within the same data split.

2.3 Target Variable

The original readmitted variable contained three categories: <30, >30, and NO. For this project, the outcome was converted into a binary variable called readmit_30. Encounters recorded as <30 were assigned a value of 1, representing readmission within 30 days, while both >30 and NO were assigned a value of 0. This binary outcome was used as the target variable for both Logistic Regression and XGBoost modeling.

3. Data Cleaning and Preparation

The initial cleaning process focused on identifying missing and invalid values. Several variables used "?" to represent missing information, so these values were converted to NaN before further processing. The highest levels of missingness were found in weight (96.86%), max_glu_serum (94.75%), A1Cresult (83.28%), medical_specialty (49.08%), and payer_code (39.56%). Because these variables contained substantial missing information and were not essential to the main research question, they were removed from the analytical dataset. Race had approximately 2.23% missing values, while missingness in the three diagnosis fields was relatively low.

Demographic and diagnosis variables were also standardized before modeling. The three Unknown/Invalid gender observations were converted to missing values, while missing categorical values were later handled during preprocessing. The original diagnosis codes in diag_1, diag_2, and diag_3 were simplified into broader clinical categories, including Diabetes, Circulatory, Respiratory, Digestive, Genitourinary, Injury, Musculoskeletal, Neoplasms, and Other. This reduced the complexity of the raw diagnosis codes while retaining clinically meaningful information.

For model preparation, categorical predictors were imputed using a "Missing" category and converted to numerical form using one-hot encoding. Numerical variables were standardized using StandardScaler. The final predictor set included demographic characteristics, hospitalization measures, prior healthcare utilization, diagnosis categories, and medication-related variables. These preprocessing steps were incorporated into a scikit-learn pipeline so that the same transformations were applied consistently during both model training and testing.

4. Patient-Level Train/ Test Split

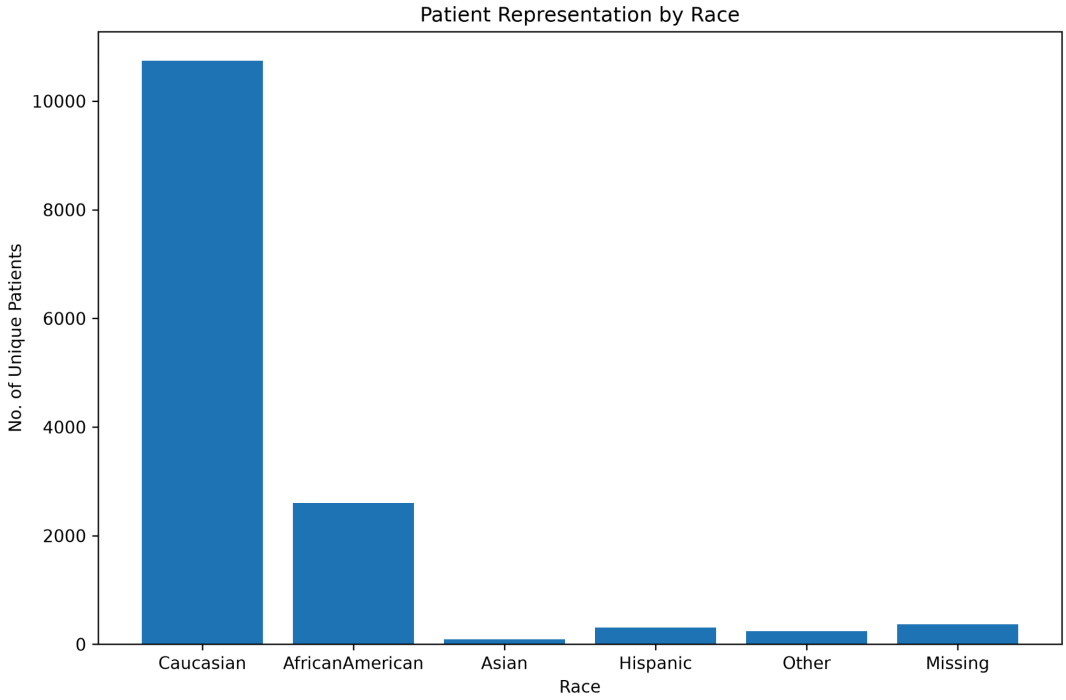
Because the dataset contained repeated hospital encounters for some patients, the training and test sets were created using patient_nbr rather than randomly splitting individual encounters. A GroupShuffleSplit with an 80/20 split and random state of 42 was used so that all encounters belonging to the same patient remained within a single dataset. This prevented information from the same patient from appearing in both training and testing and reduced the risk of data leakage. The final training set contained 81,613 encounters from 57,214 unique patients, while the test set contained 20,153 encounters from 14,304 unique patients, with zero patient overlap between the two sets.

The proportion of 30-day readmissions remained similar across the two datasets. The training set contained 9,206 positive readmission cases, representing approximately 11.28% of training encounters, while the test set contained 2,151 positive cases, representing approximately 10.67% of test encounters. This similarity helped ensure that the held-out test set provided a reasonable representation of the original outcome distribution while remaining completely independent at the patient level.

5. Exploratory Analysis

Exploratory analysis was conducted to examine the distribution of demographic groups and 30-day readmission rates before model development. The dataset was predominantly Caucasian, with 76,099 encounters, followed by 19,210 African American, 2,037 Hispanic, 1,506 Other, and 641 Asian encounters; race was missing for 2,273 encounters. Gender representation was more balanced, with 54,708 female and 47,055 male encounters. Age was concentrated in the older groups, particularly ages 60–89.

Fig 1. Distribution of hospital encounters across racial groups.



Thirty-day readmission rates were fairly similar across the two major gender groups, at 11.25% for females and 11.06% for males. Across racial groups, readmission rates ranged from 9.63% for Other to 11.29% for Caucasian patients, while the Asian and Hispanic groups had rates of 10.14% and 10.41%, respectively. Age showed somewhat greater variation, with lower readmission among the youngest patients and rates generally around 10–12% among older age groups.

These patterns showed that the demographic groups differed substantially in representation even when their observed readmission rates were relatively similar. This was especially important for the later

subgroup analysis because smaller groups contained fewer patients and fewer positive readmission cases, which could make performance estimates less stable.

6. Machine Learning Methodology

Two supervised classification models were developed to predict 30-day hospital readmission: Logistic Regression and XGBoost. Logistic Regression was used as an interpretable baseline model, while XGBoost was selected as the nonlinear model because it can capture more complex relationships among demographic, hospitalization, diagnosis, medication, and prior-utilization variables. To address class imbalance, Logistic Regression was trained using balanced class weights, while XGBoost used a scale_pos_weight value based on the ratio of non-readmitted to readmitted cases in the training data.

Both models were evaluated on the same patient-separated test dataset using AUROC, AUPRC, precision, recall, F1-score, and confusion matrices. AUROC was used to evaluate overall discrimination between readmitted and non-readmitted encounters, while AUPRC was included because the positive 30-day readmission class represented a relatively small proportion of the data. Recall was also important because it measured the proportion of actual 30-day readmissions correctly identified by the model, while precision and F1-score helped evaluate the trade-off between identifying positive cases and generating false-positive predictions.

7. Model Results

Both models showed modest ability to distinguish encounters followed by 30-day readmission from those that were not. Logistic Regression achieved an AUROC of 0.632, AUPRC of 0.180, precision of 0.160, recall of 0.497, and F1-score of 0.242. XGBoost performed slightly better in discrimination and recall, achieving an AUROC of 0.637, AUPRC of 0.187, recall of 0.525, precision of 0.157, and F1-score of 0.242.

Table 1. Overall Model Performance

Model	AUROC	AUPRC	Precision	Recall	F1
Logistic Regression	0.632	0.180	0.160	0.497	0.242
XGBoost	**0.637**	**0.187**	0.157	**0.525**	0.242

Fig 2. ROC curve comparison of Logistic Regression and XGBoost.

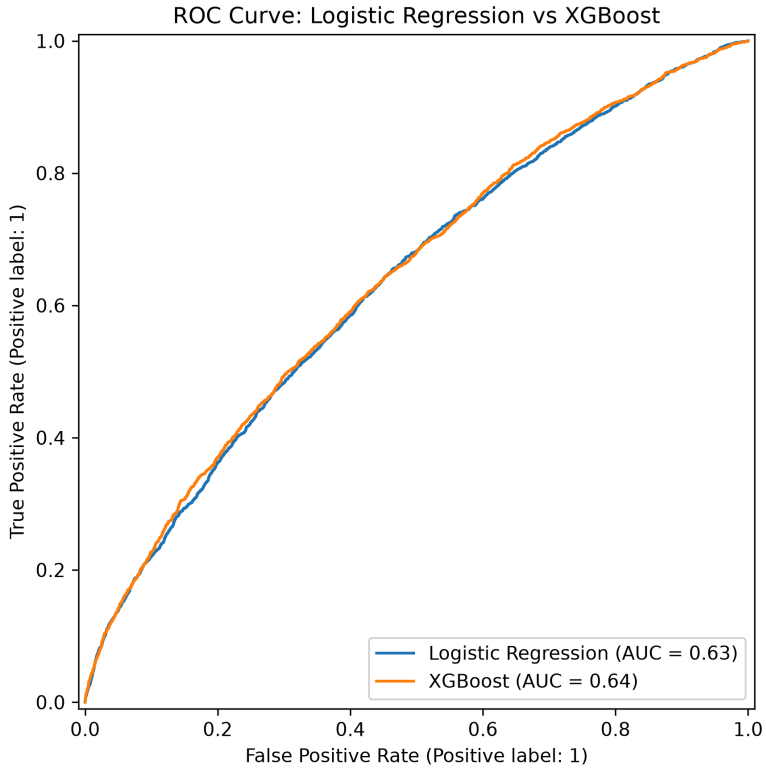
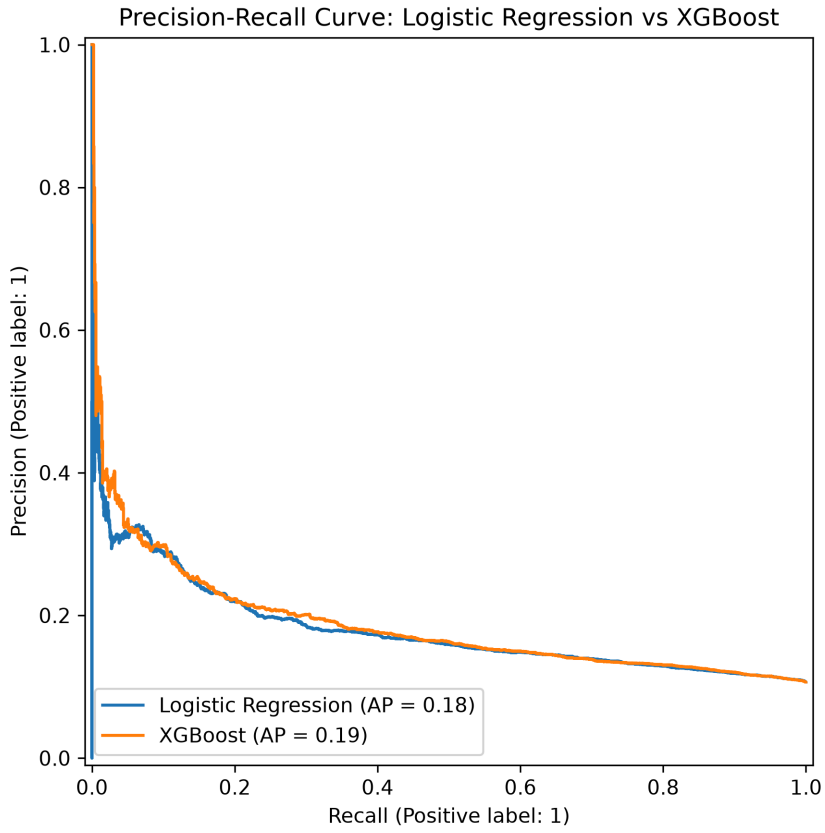


Fig 3. Precision–Recall curve comparison of Logistic Regression and XGBoost.



XGBoost was selected for the demographic subgroup analysis because it produced the highest AUROC, AUPRC, and recall while maintaining an F1-score comparable to Logistic Regression. Its confusion matrix showed 11,951 true negatives, 6,051 false positives, 1,022 false negatives, and 1,129 true positives. Although the improvement over Logistic Regression was small, XGBoost identified a greater proportion of actual 30-day readmissions and was therefore used as the primary model for subsequent subgroup comparisons.

8. Demographic Subgroup Performance

The selected XGBoost model was evaluated separately across race, gender, and age groups to determine whether overall performance was consistent across demographic populations. Subgroup evaluation included the number of encounters and positive readmission cases together with AUROC, recall, precision, and F1-score.

8.1 Performance by Race

Performance varied more across racial groups than across gender groups. Caucasian and African American patients showed nearly identical recall values of 0.526 and 0.527, respectively. Hispanic patients had the highest recall at 0.548, while the Asian subgroup had the lowest recall at 0.368 and the lowest AUROC at 0.608. The Asian test subgroup was also very small, containing only 125 encounters, 92 unique patients, and 19 positive readmission cases, compared with 15,089 Caucasian encounters and 1,626 positive cases.

Table 2. Performance: Race

Race	Encounters	Positive Cases	AUROC	Recall	Precision	F1
Caucasian	15,089	1,626	0.633	0.526	0.155	0.239
African American	3,847	425	0.636	0.527	0.167	0.253
Asian	125	19	0.608	0.368	0.200	0.259
Hispanic	393	31	0.671	0.548	0.145	0.230
Other	289	29	0.697	0.517	0.217	0.306

8.2 Performance by Gender

Gender performance was relatively consistent. Female patients achieved an AUROC of 0.632 and recall of 0.538, while male patients achieved an AUROC of 0.642 and recall of 0.509. Precision and F1-score

were also nearly identical between the two groups, suggesting that the model did not show a large gender-related performance difference in this test dataset.

8.3 Performance by Age Group

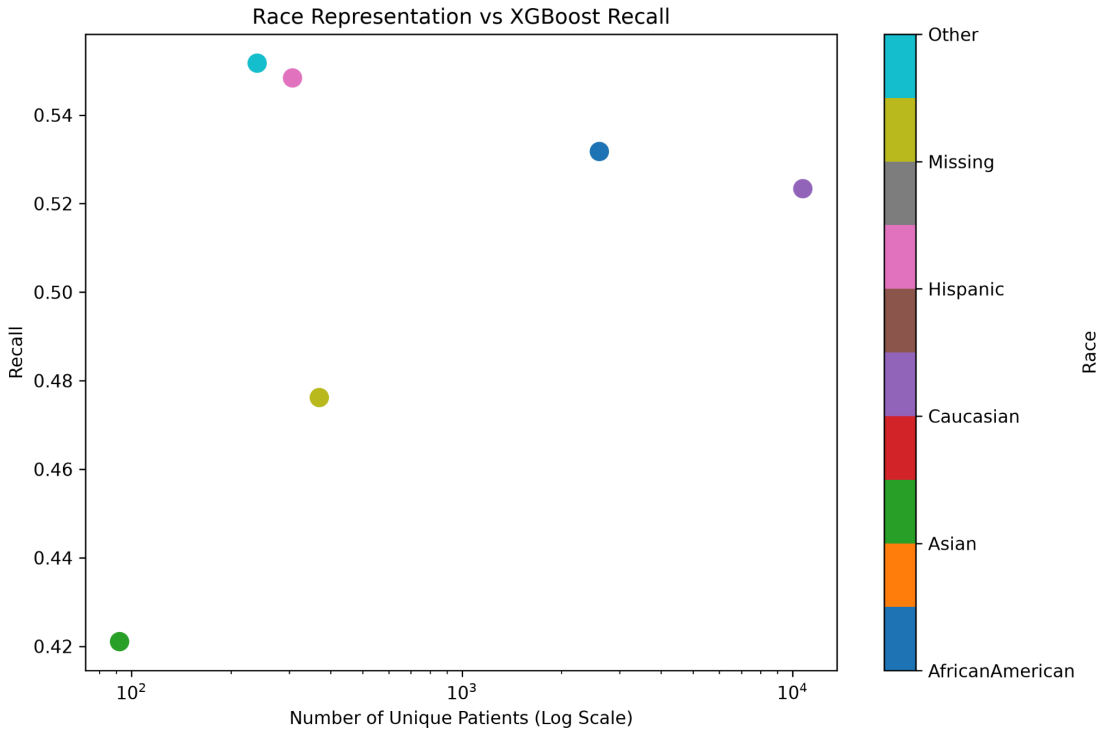
Age-group results showed a different pattern. AUROC declined from 0.686 among patients aged 0–39 to 0.601 among patients aged 80–100. However, recall did not decline in the same way: the 60–79 and 80–100 groups had recall values of approximately 0.556 and 0.554, compared with 0.504 for ages 0–39 and 0.448 for ages 40–59. These results demonstrate that subgroup performance depended on the metric being considered and should not be evaluated using a single measure alone.

9. Representation and Weighting Analysis

9.1 Demographic Representation

Racial groups were unevenly represented in the training data. Caucasian patients accounted for the largest number of encounters (61,010), followed by African American patients (15,363). In contrast, Hispanic, Other, and Asian groups had substantially fewer encounters, with the Asian group containing only 516 training encounters. In the test set, the Asian subgroup also contained only 92 unique patients and 19 positive readmission cases. The lower representation of this subgroup coincided with the lowest race-specific recall of 0.368, suggesting that subgroup sample size may influence the stability of performance estimates. However, these results do not demonstrate that lower representation directly caused poorer model performance.

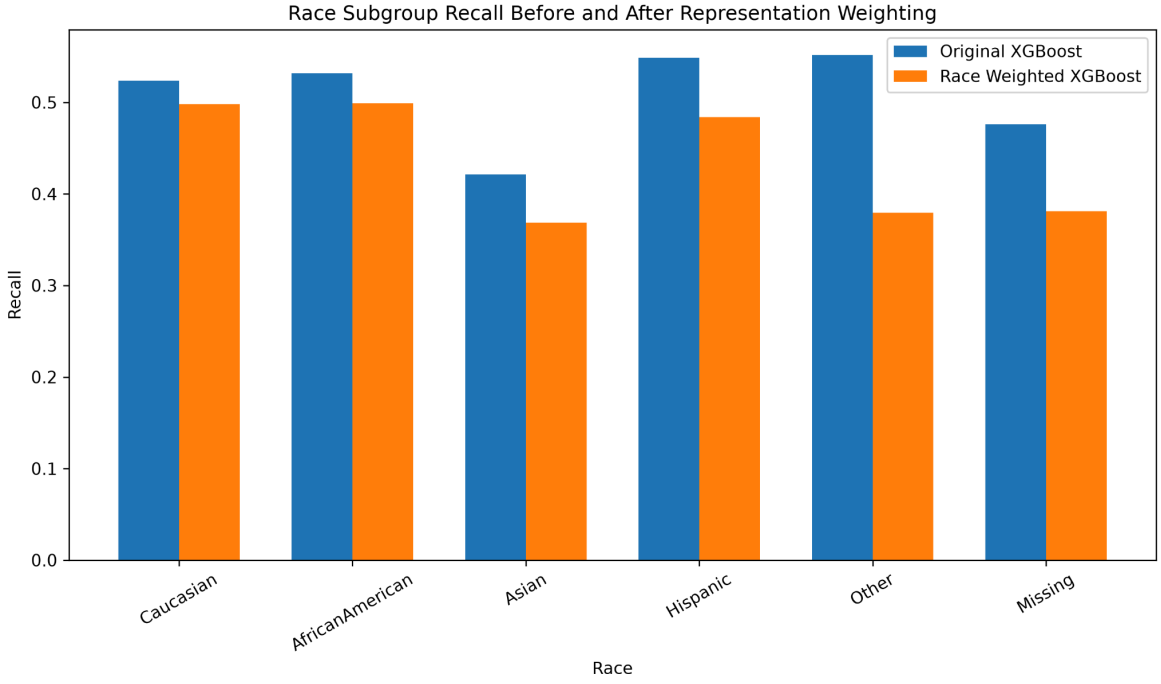
Fig 4. Relationship between racial group representation and XGBoost recall.



9.2 Race-Based Sample Weighting

To examine whether unequal racial representation influenced model performance, a second XGBoost model was trained using race-based sample weights. This approach gave greater training influence to less-represented racial groups. After weighting, however, the Asian subgroup's recall remained unchanged at 0.368, although its precision increased from 0.200 to 0.233 and F1-score increased from 0.259 to 0.286. Recall decreased for several other racial groups. At the overall level, weighting reduced AUROC from 0.637 to 0.626, AUPRC from 0.187 to 0.181, and recall from 0.525 to 0.497, while precision increased slightly from 0.157 to 0.161. These findings indicate that balancing racial representation through sample weighting alone did not consistently improve subgroup or overall performance, suggesting that observed demographic differences may involve factors beyond representation alone.

Fig 5. Race subgroup recall before and after sample weighting.

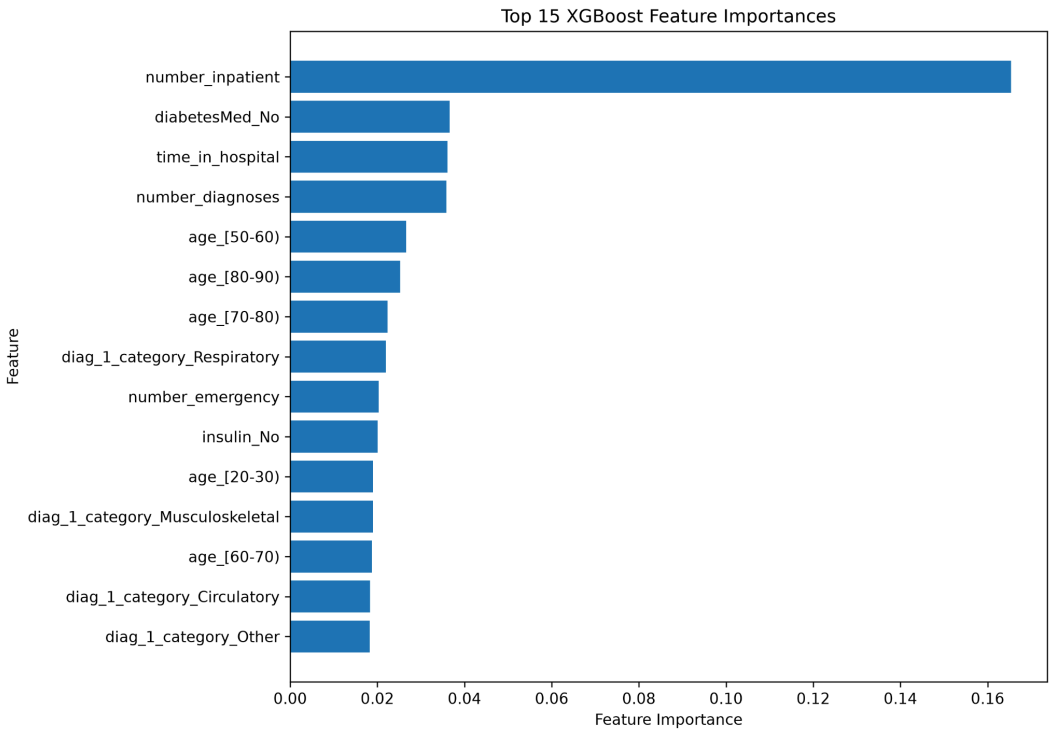


10. Feature Importance

Feature importance from the XGBoost model was examined to identify which predictors contributed most strongly to readmission predictions. The most influential feature was number_inpatient, representing the number of previous inpatient visits, with substantially greater importance than the remaining variables. Other important predictors included time in hospital, number of diagnoses, diabetes medication status, insulin use, emergency visits, age categories, and diagnosis categories. These results suggest that previous healthcare utilization and characteristics of the current hospitalization played an important role in predicting 30-day readmission.

Among demographic variables, several age categories appeared within the most important features, including ages 50–60, 70–80, and 80–90. race_Caucasian also appeared among the top 15 features, although its importance was much smaller than that of prior inpatient utilization. Feature importance should not be interpreted as evidence that a variable causes readmission; instead, it indicates how much the trained XGBoost model relied on that feature when generating predictions. Overall, the results show that the model relied on a combination of prior healthcare use, clinical information, treatment-related variables, and demographic characteristics rather than on demographic information alone.

Fig 6. Top feature importance values from the XGBoost readmission model.



11. Discussion

The results showed that both Logistic Regression and XGBoost had only moderate predictive performance for 30-day hospital readmission, with XGBoost performing slightly better overall. Its AUROC of 0.637 and AUPRC of 0.187 indicate that predicting short-term readmission remains difficult using the available variables. Although XGBoost achieved a somewhat higher recall than Logistic Regression, the improvement was small, suggesting that the added nonlinear complexity did not produce a major increase in predictive accuracy.

Demographic subgroup analysis revealed that performance was not completely consistent across all groups. Gender differences were relatively small, while larger variation appeared across race and age groups. The Asian subgroup had the lowest race-specific recall and was also the least represented racial group, but its very small number of positive cases makes this estimate less stable. Age groups showed declining AUROC among older patients, while recall remained relatively high, demonstrating that subgroup conclusions can change depending on the performance metric considered. The race-based weighting experiment did not consistently improve subgroup performance and reduced several overall metrics, indicating that unequal representation alone does not fully explain the observed differences. These findings therefore represent subgroup performance differences rather than direct evidence of model bias, and they highlight the importance of considering representation, sample size, and multiple evaluation metrics when assessing machine-learning models across demographic populations.

12. Limitations

Several limitations should be considered when interpreting the results of this project. First, the dataset contains hospital encounters from 1999–2008, so clinical practices, treatment patterns, and hospital systems may differ from current healthcare settings. The dataset also focuses on patients with diabetes, which limits the ability to generalize the findings to all hospitalized populations. In addition, the overall predictive performance of both models was modest, indicating that important factors influencing 30-day readmission may not be fully represented by the available variables.

Subgroup results should also be interpreted cautiously because demographic groups were not equally represented. Some racial groups, particularly Asian and Hispanic patients, contained relatively few patients and positive readmission cases, making their performance estimates less stable than those of larger groups. The analysis evaluated differences in AUROC, recall, precision, and F1-score, but these differences do not by themselves establish algorithmic bias or causal effects of demographic representation. Finally, the race-based weighting experiment represented only one approach to addressing representation differences; other modeling strategies, datasets, or fairness evaluation methods could produce different results.

13. Ethical and Practical Considerations

Machine-learning models used in healthcare can influence how patients are identified for additional monitoring, follow-up, or care coordination. For this reason, evaluating only overall predictive performance is not sufficient. A model may appear acceptable at the population level while performing differently across demographic groups. In this project, subgroup performance was therefore examined across race, gender, and age to better understand whether prediction quality was consistent for different patient populations.

The observed demographic differences should not automatically be interpreted as evidence of unfairness or discrimination. Differences in sample size, readmission prevalence, clinical characteristics, and representation may all affect subgroup metrics. In particular, estimates for small groups can be unstable when only a limited number of positive cases are available. Any real-world use of a readmission model should therefore include continued subgroup monitoring, validation on more recent and diverse populations, and human clinical oversight rather than relying on model predictions alone.

14. Conclusion

This project evaluated whether machine-learning models for 30-day hospital readmission performed consistently across demographic groups using the UCI Diabetes 130-US Hospitals dataset. Logistic Regression and XGBoost were compared, with XGBoost achieving slightly better overall performance, including an AUROC of 0.637, AUPRC of 0.187, and recall of 0.525. Subgroup analysis showed relatively small performance differences by gender, while greater variation was observed across race and age groups. In particular, the Asian subgroup had the lowest race-specific recall and was also the least represented racial group in the test data.

The representation-weighting experiment showed that increasing the training influence of smaller racial groups did not consistently improve subgroup performance and slightly reduced overall AUROC, AUPRC, and recall. Therefore, the observed demographic differences cannot be explained by representation alone. Overall, the findings highlight the importance of evaluating healthcare prediction models not only by overall accuracy but also by demographic subgroup performance, sample size, and multiple evaluation metrics. Further research using larger, more recent, and more diverse datasets would be valuable for developing more reliable and consistent readmission prediction models.

Reference

Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). Diabetes 130-US Hospitals for Years 1999–2008 [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J